EEE-452: ANTENNA ENGINEERING PROJECT ANALYSIS REPORT

YAGI-UDA ANTENNA

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Recap of Design Proposal: As a team, we proposed a Yagi-Uda Antenna, a linearly (horizontally) polarized antenna for our term project. We considered two different types of Yagi-Uda Antenna in our proposal, as 3-D and Quasi, to choose one of them during the process and avoid manufacturing and pattern problems that we may encounter. We aim to design the antenna to operate at 3 GHz (wavelength of 10 cm), achieve a matching impedance, and consequently, $|S_{11}| < -10$ dB, with an expected HPBW of 40–60°. The 3-D method is expected to have a maximum directivity of 8-10 dBi. The quasi method is expected to have a maximum directivity of 3-5 dBi. We will use a 50Ω SMA connector and a 6-element topology.

Design Process and Obtained Results: We designed the two Yagi-Uda configurations mentioned above using CST.

- **Quasi Yagi-Uda Design Process**

Firstly, the Quasi Yagi-Uda is designed on CST, following the dimensions and optimisations used by Berkay Atabay [1]. After the pattern was obtained in a similar shape to the expected results of Atabay [1],[2], the next step in the design process was matching. The resonant frequency is found to be dependent mostly on the driver length. Therefore, the driver length was optimised to bring the resonant frequency to 3 GHz. As explained in the proposal, we expected a directivity of 3-5 dBi and an HPBW of 40-60 degrees. We obtained a directivity of 5.3 dBi, and the HPBWs are found to be 60° and 150°. The final design features 1 reflector, 1 driven element, and 5 directors. Overall, we used a 7-element topology, as opposed to the proposed 6-element topology.

- **3-D Yagi-Uda Design Process**

Firstly, the 3-D Yagi-Uda antenna is designed on CST, following the instructions given in NBS Technical Notes [3]. This design has 6 elements, namely: 1 reflector, 1 driven element (dipole) and 4 directors as previously mentioned in the proposal. After acquiring the pattern cuts as the same as the NBS Technical notes, by using optimizer and parameter sweep techniques, the matching was done [3]. Therefore, the driver length was optimised to bring the resonant frequency to 3 GHz. As explained in the proposal, we expected a directivity of 8-10 dBi and an HPBW of 40-60 degrees. We have acquired a -18.15 dB matching at 3 GHz. The resonant frequency is mostly dependent on the driver length. But the parasitic effects of the Yagi-Uda shift the frequency to the left, i.e, to lower frequencies. To compensate for this, the wavelength was decreased, changing the overall dimensions, to get a better matching at 3 GHz. The directivity of the antenna is found to be 11.6 dBi, which is slightly higher than expected. This is due to the fact that we have used PEC for simulating the antenna's behavior. The HPBWs are found to be at 44.2°, and 51.8°.

Bill of Materials (BoM): The following materials are planned for use in the manufacturing process.

Method 1: 3-D Antenna

1) Director, Reflector, Driven Elements
- Bicycle wheel spokes (Aluminum)
- Umbrella frames
- Clothes hanger

2) Boom
- Wand-like cat toy (broken)
- Wooden plate from old projects

Method 2: Quasi Antenna

1) Director, Reflector, Driven Elements
- Milk box
- Metal beverage can
- Aluminium foil from old projects

2) Boom
- Old cardboard box
- Cellulose acetate sheets from old spiral notebooks

Simulation Results of Proposed Design:

- **Quasi Yagi-Uda Simulations**

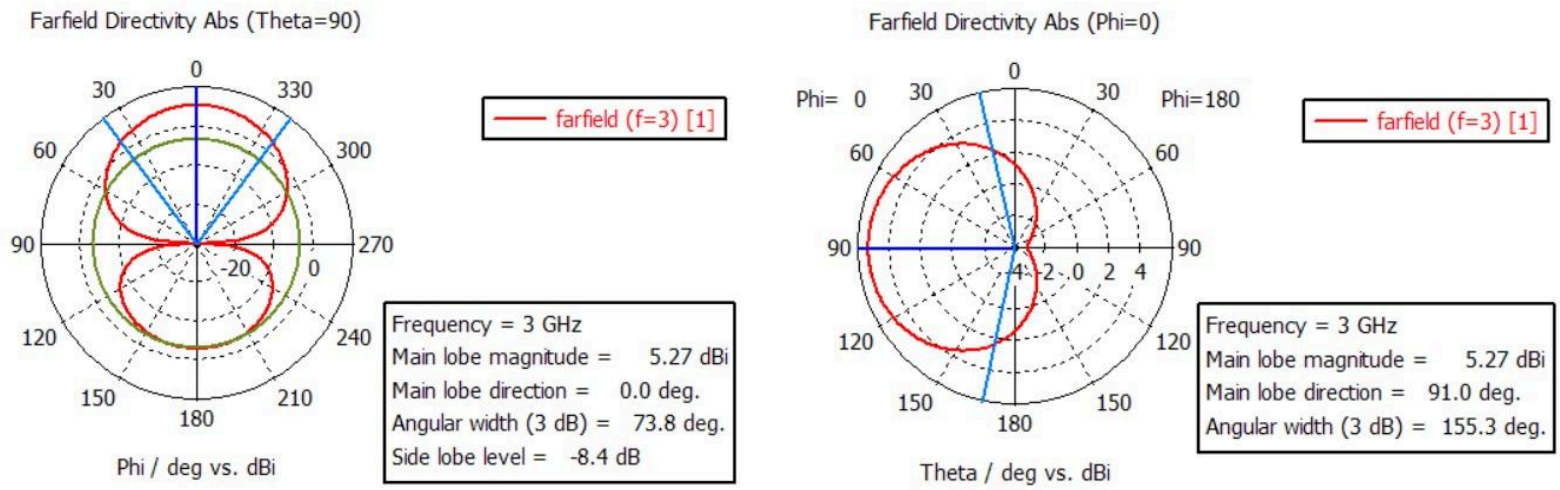


Figure 1.1: Field Patterns of Quasi Yagi-Uda

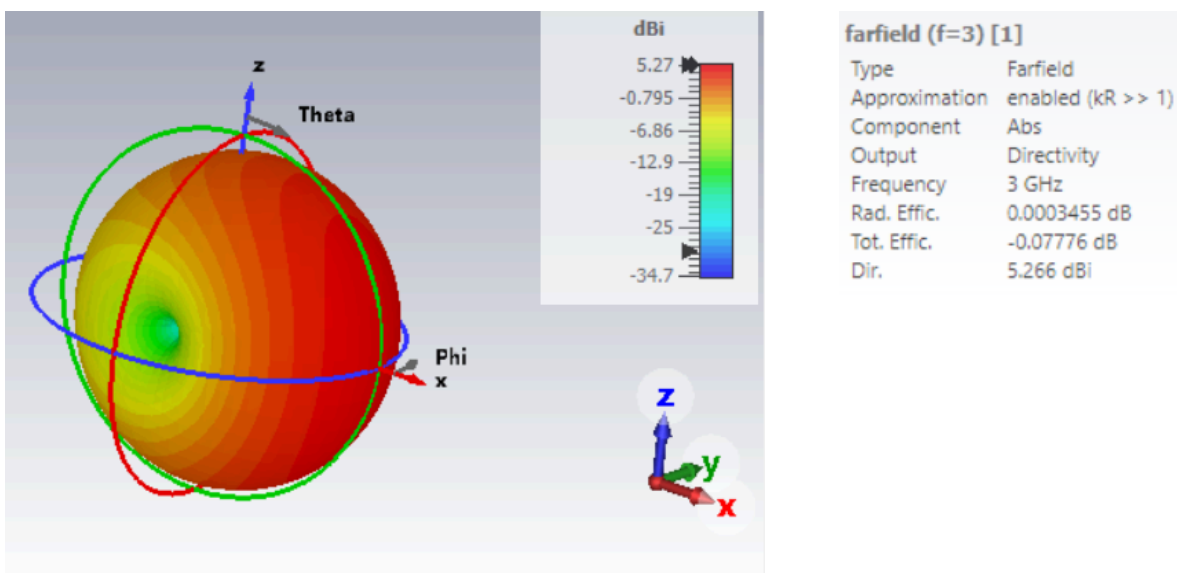


Figure 1.2: Directivity of Quasi Yagi-Uda

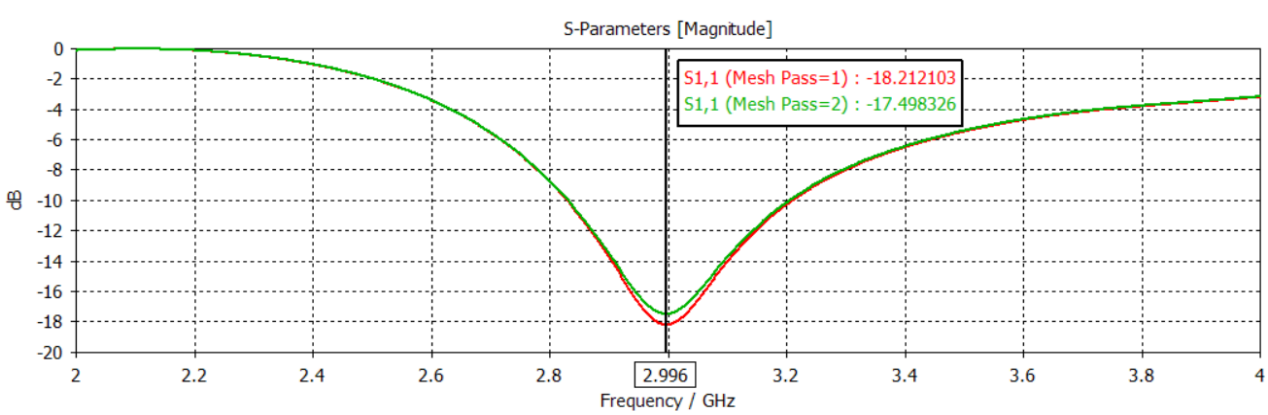


Figure 1.3: S_{11} parameter of Quasi Yagi-Uda

- **3-D Yagi-Uda Simulations**

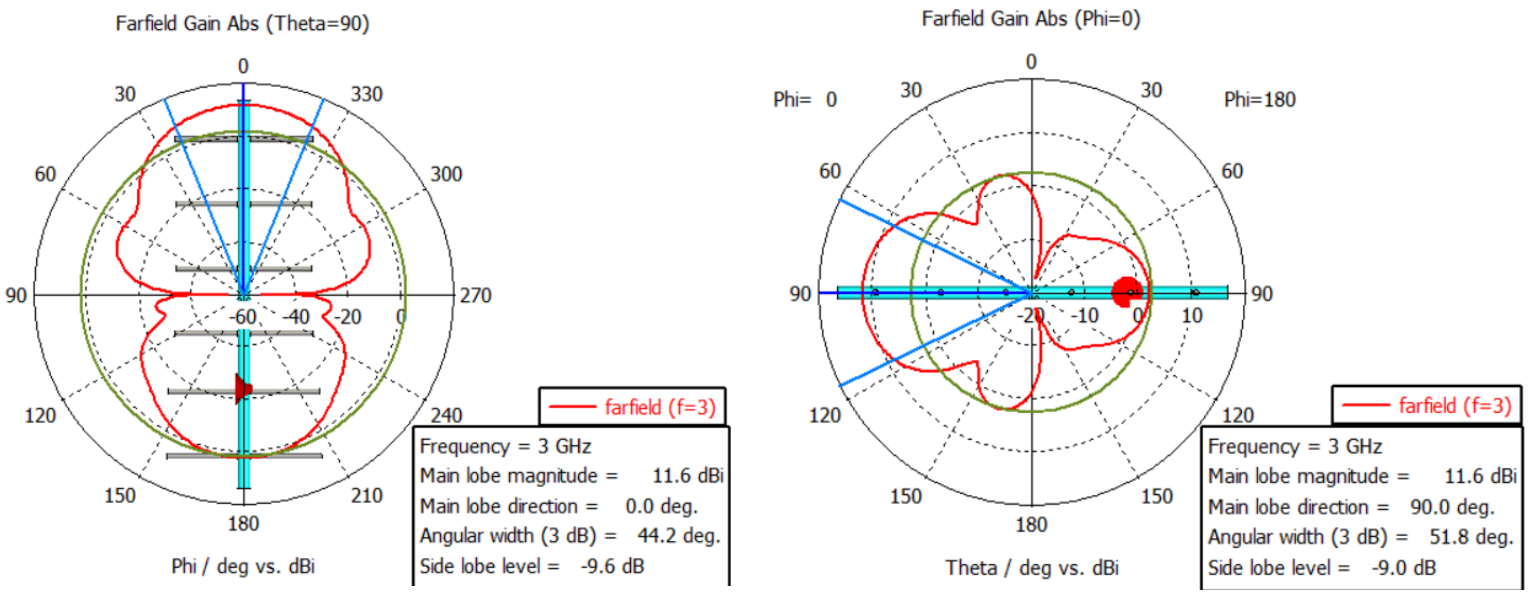


Figure 2.1: Field Patterns of 3-D Yagi-Uda

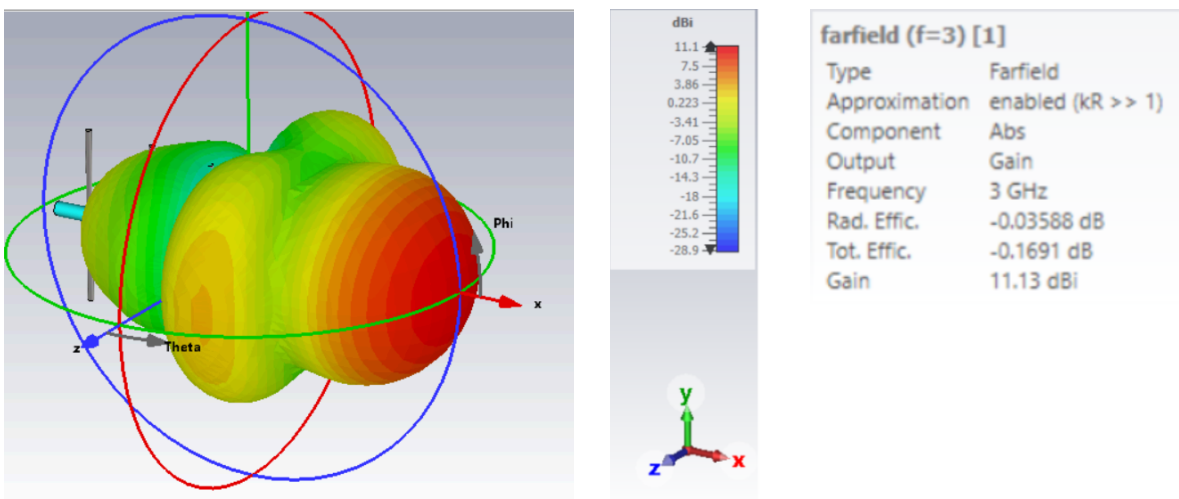


Figure 2.2: Directivity of 3-D Yagi-Uda

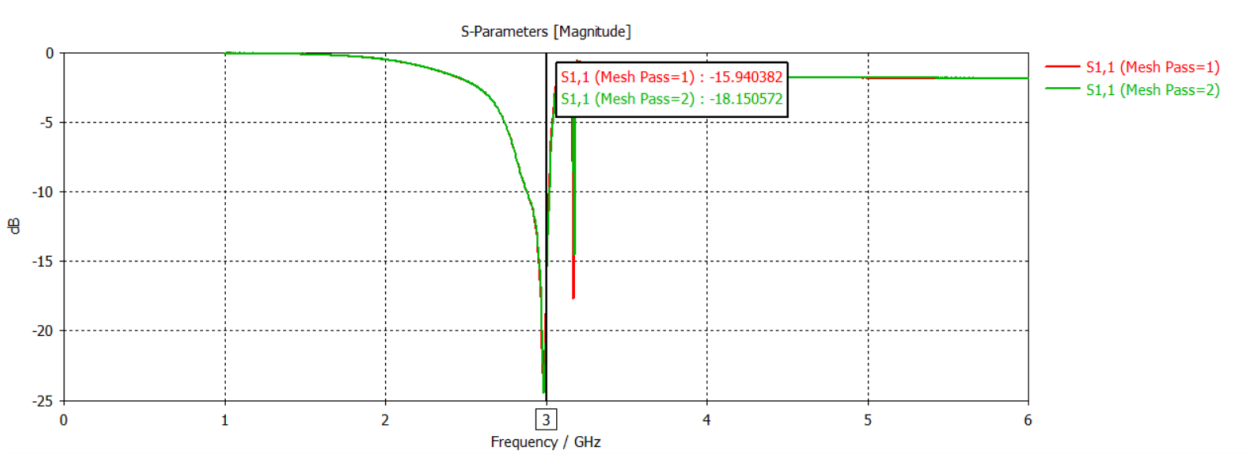


Figure 2.3: S_{11} parameter of 3-D Yagi Uda

Simulation Parameters, Dimensions, and Material Properties:

- **Quasi Yagi-Uda Simulation Parameters:**

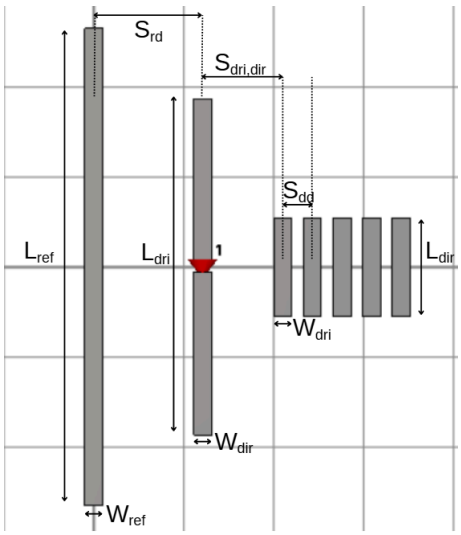
Geometry of the Antenna	Dimensions	Simulation Parameters
	$\lambda = 100mm$ $S_{rd} = 0.121\lambda$ $S_{dri,dir} = 0.089\lambda$ $S_{dir,dir} = 0.033\lambda$ $L_{ref} = 0.53\lambda$ $L_{dri} = 0.37\lambda$ $L_{dir} = 0.11\lambda$ $W_{dir} = 0.02\lambda$ $W_{ref} = 0.02\lambda$ $W_{dri} = 0.02\lambda$ Thickness = 0.01λ $\epsilon_{r,dielectric} = 2.7$	- Adaptive Mesh Refinement - [2,4]GHz Frequency Range - %75 CPU, %80 RAM(16GB) Usage - 340k Meshes - 148 seconds run time for meshing and simulating only the given geometry.

Figure 3.1: Quasi Yagi-Uda Parameters

- **3-D Yagi-Uda Simulation Parameters:**

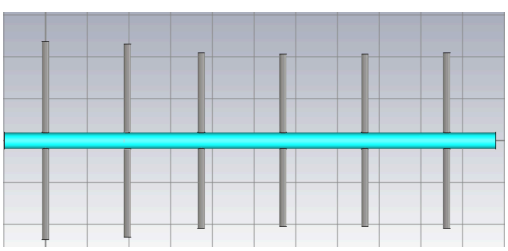
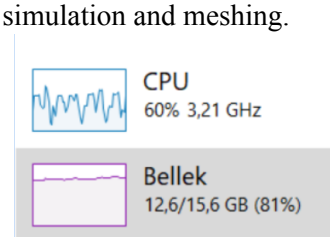
Geometry of the Antenna	Dimensions	Simulation Parameters
	$S_{rd} = 0.2\lambda$ $S_{dri,dir} = 0.18\lambda$ $S_{dir,dir} = 0.2\lambda$ $L_{ref} = 0.482\lambda$ $L_{dri} = 0.471\lambda$ $L_{dir1,4} = 0.428\lambda$ $\lambda = 98mm$ $L_{dir2,3} = 0.420\lambda$ $D_{dir} = 0.0085\lambda$ $D_{ref} = 0.0085\lambda$ $D_{dri} = 0.0085\lambda$ $\epsilon_{r,boom} = 1.74$	-Adaptive Mesh Refinement -[1,6]GHz Frequency Range -%60 CPU, %81 RAM (16GB) Usage -387k Meshes -15 minutes and 32 seconds run time for simulation and meshing. 

Figure 3.2: 3-D Yagi-Uda Parameters

Progress: No progress has been made in the manufacturing process.

REFERENCES:

[1] Atabay, B., "*Design and Realization a Quasi Yagi Antenna Array and Its Feed System Printed on a Ceramic Substrate for X-Band Radar Applications*", Middle East Technical University, Ankara, Dec.2019.[Online]. Available: https://etd.lib.metu.edu.tr/upload/12624988/index.pdf

[2] Constantinescu, C., Pacurar, C., Giurgiuman, A., Munteanu, C., Andreica, S., & Gliga, M., *"High Gain Improved Planar Yagi Uda Antenna for 2.4 GHz Applications and Its Influence on Human Tissues*," Applied Sciences, May 2023. [Online]. Available: https://www.mdpi.com/2076-3417/13/11/6678

[3] Peter P. Viezbicke, *Yagi Antenna Design*, NBS Technical Note 688, December 1976.